

# Engineering Mathematics - 1

[Application oriented syllabus]

Unit : 1

## ★ Unit : 1. differential Calculus.

- 1) Rolle's theorem.
- 2) Mean Value Theorem.
- 3) Taylor's Series, Maclaurin's Series
- 4) Expansion of a func<sup>n</sup> using standard expansions
- 5) Indeterminate forms.
- 6) L' Hospital's Rule.

## ★ Unit : 2. Fourier Series

- 1) Definition of full-range Fourier series.
- 2) Half - range Fourier series.
- 3) Harmonic analysis
- 4) Application to the problem in engineering.

## ★ Unit : 3. Partial Differentiation.

- 1) Introduction to the func<sup>n</sup> of several variables
- 2) Partial derivatives
- 3) Euler's Theorem.
- 4) Partial derivative of composite func<sup>n</sup>.
- 5) Change of independent variables.

## ★ Unit : 4. Application of Partial Differentiation.

- 1) Jacobian and its applications.
- 2) Errors and approximation.
- 3) Maxima and minima.

## ★ Unit : 5. Linear Algebra

Matrices & System of linear equations.

- 1) Rank of matrix.
- 2) System of linear equations

- 1) Linear dependence & independence.
- 2) Linear and orthogonal transformation.
- 3) Application to the problem in engineering.

- Unit : 6. Linear Algebra
- 1) Eigen value & eigen vector
  - 2) Diagonalization
  - 3) Diagonalization of matrix.

## Unit : 1. Differentiation Calculus.

Every polynomial func<sup>n</sup> is conti & diff. except  $\lfloor x$   
Every trigonometric func<sup>n</sup> is conti & diff.

### 1) Rolle's Theorem :-

- Rolle's The<sup>m</sup> has 3 conditions. (3 = they ex
- i)  $f(x)$  is conti on  $[a, b]$
- ii)  $f(x)$  is diff on  $(a, b)$
- iii)  $f(a) = f(b)$  then  $\exists c \in (a, b)$  such that  $\boxed{f'(c) = 0}$

Ex : i) Verify Rolle's theorem for  $f(x) = x^3$  on  $[3, 7]$

- i)  $f(x) = x^3$  is conti on  $[3, 7]$
- ii)  $f(x) = x^3$  is diff on  $(3, 7)$
- iii)  $f(a) = (3)^3 = 9$   
 $f(b) = (7)^3 = 49$  }  $9 \neq 49$   
i.e.  $f(a) \neq f(b)$

$\Rightarrow$  Rolle's th<sup>m</sup> is not verified

### 2) Verify Rolle's theorem for $f(x) = x^2 - 10x + 16$ on $[3, 7]$

- i)  $f(x) = x^2 - 10x + 16$  is conti on  $[3, 7]$
- ii)  $f(x) = x^2 - 10x + 16$  is diff on  $(3, 7)$
- iii)  $a = 3, b = 7$

$$f(a) = (3)^2 - 10(3) + 16 \Rightarrow 9 - 30 + 16 \Rightarrow -5$$

$$f(b) = (7)^2 - 10(7) + 16 \Rightarrow 49 - 70 + 16 \Rightarrow -5$$

$$f(a) = f(b) \therefore f(0) = [3, 7]$$

such that  $f'(c) = 0$ .

$$f'(x) = 2x - 10$$

$$f'(c) = 2c - 10 \Rightarrow c = \frac{10}{2} \Rightarrow \boxed{c = 5}$$

3) Verify Rolle's th<sup>m</sup> for  $x^3 - 12x$  in  $[0, 2\sqrt{3}]$ .  
func is polynomial.

→ iii)  $f(a) = f(b)$

$f(a) = f(0) = -12(0) \Rightarrow 0 - 12 = 0$

$f(b) = (2\sqrt{3})^3 - 12(2\sqrt{3}) \Rightarrow f(2\sqrt{3})$

$24\sqrt{3} - 24\sqrt{3} = 0$

$f(a) = f(b) \quad \therefore f(0) = f(2\sqrt{3})$

$\Rightarrow \exists c \in (0, 2\sqrt{3})$

such that  $f'(c) = 0$

$f'(x) = 3x^2 - 12$

$f'(c) = 3c^2 - 12$

$3c^2 - 12 = 0$

$3c^2 - 12 \Rightarrow c^2 = 12/3 \Rightarrow c^2 = 4 \Rightarrow c = 2, -2$

$c = 2$  is lies on  $[0, 2\sqrt{3}]$

Rolle's th<sup>m</sup> verified.

i)  $f(x) = x^3 - 12x$  is conti on  $[0, 2\sqrt{3}]$

ii)  $f(x) = x^3 - 12x$  is diff on  $(0, 2\sqrt{3})$

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4) Verify Rolle's theorem for  $f(x) = \sin x - \sin 2x$  on the interval  $[0, \pi]$ .

→ i)  $f(x)$  is conti on  $[0, \pi]$

ii)  $f(x)$  is diff on  $(0, \pi)$

iii)  $a = 0, b = \pi$

$f(a) = f(0) = f(0) = \sin 0 - \sin 2(0)$

$= 0 - 0 = 0$

$f(b) = f(\pi) = \sin \pi - \sin 2\pi = 0 - 0 = 0$

$f(a) = f(b)$

$\therefore \exists f(a) = f(\pi)$

$\exists c \in (0, \pi)$  such that  $f'(c) = 0$

$f'(c) = 0$

$f(x) = \sin x - \sin 2x$

$f'(x) = \cos x - 2\cos 2x$

$f'(c) = \cos c - 2\cos 2c = 0$

$f'(c) = \cos c - 2[2\cos^2 c - 1] = 0$

$\{ \cos 2c = 2\cos^2 c - 1 \}$

$= \cos c - 4\cos^2 c + 2 = 0$

$\therefore 4\cos^2 c - \cos c - 2 = 0$

$ax^2 + bx + c = 0$

$a = 4, b = -1, c = -2$

$\frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \Rightarrow \frac{1 \pm \sqrt{(-1)^2 - 4(4)(-2)}}{2(4)}$

$\Rightarrow \frac{1 \pm \sqrt{33}}{8}$

$\therefore \cos c = \frac{1 \pm \sqrt{33}}{8}$

$c = \cos^{-1} \left[ \frac{1 \pm \sqrt{33}}{8} \right] \Rightarrow \cos^{-1} [0.82] \Rightarrow 34.5$

$\therefore c$  is lies on  $[0, \pi] \Rightarrow$  Rolle's Th<sup>m</sup> verified

VIM<sup>8</sup>

Lagrange's Mean Value Theorem (LMVT).

Conditions:

i)  $f(x)$  is conti on  $[a, b]$

ii)  $f(x)$  is diff on  $(a, b)$

iii)  $f'(c) = f(b) - f(a)$

$b - a$

Suppose:

i)  $f(x)$  is conti on  $[a, b]$

ii)  $f(x)$  is diff on  $(a, b)$

then  $\exists$  atleast one point  $c$   
 $\exists c \in (a, b)$   
 such that  $f'(c) = \frac{f(b) - f(a)}{b - a}$

Ex: 1) Verify LMVT for  $f(x) = x^2$  in  $[1, 5]$ .

$\rightarrow$  i) Every polynomial func<sup>n</sup> is conti on  $[a, b]$ .

$\therefore f(x) = x^2$  is conti on  $[1, 5]$ .

ii) Every polynomial func<sup>n</sup> is diff on  $(a, b)$ .

$f(x) = x^2$  is diff on  $(1, 5)$

iii)  $f(a) = 1$   $f(b) = 5$

$f'(c) = \frac{f(b) - f(a)}{b - a}$   $\Rightarrow$   $\text{---} \otimes$

$f(b) - f(a) = 25 - 1 = 24$

$\frac{f(b) - f(a)}{b - a} = \frac{24}{4} = 6$

$b - a$

$f'(x) = 2x$

$f'(c) = 2c$

$2c = 6$

$\therefore c = 3 \in (1, 5)$

$\Rightarrow$  LMVT verified.

Ex: 2)  $f(x) = x^3 - 1$  on  $[2, 3]$

$\rightarrow$  i)  $f(x) = x^3 - 1$  is conti on  $[2, 3]$ .

ii)  $f(x) = x^3 - 1$  is diff on  $(2, 3)$ .

iii)  $f'(c) = \frac{f(b) - f(a)}{b - a}$   $\text{---} \otimes$

$f(b) - f(a) = 3^3 - 1 = 26$

$\frac{f(b) - f(a)}{b - a} = \frac{26}{1} = 26$

$b - a$

$f(b) - f(a) = (3^3 - 1) - (2^3 - 1) = (9 - 1) - (4 - 1)$   
 $= 8 - 3 = 5$

$\therefore \frac{f(b) - f(a)}{b - a} = \frac{5}{1} = 5$

$b - a$

$f'(x) = 2x$

$f'(c) = 2c = 5$

$2c = 5 \Rightarrow c = 5/2 = 2.5$

$\therefore c = 2.5 \in [2, 3]$

$\Rightarrow$  LMVT verified.

Ex: 3)  $f(x) = \sin x$   $[0, \frac{\pi}{2}]$  LMVT.

$\rightarrow$  i)  $f(x) = \sin x$  is conti on  $[0, \frac{\pi}{2}]$ .

ii)  $f(x) = \sin x$  is diff on  $(0, \frac{\pi}{2})$ .

iii)  $f'(c) = \frac{f(b) - f(a)}{b - a}$

$b - a$

$f(a) = \sin(0) = 0$   $f(b) = \sin(\frac{\pi}{2}) = 1$   $f'(c) = \cos c$

$\cos c = \frac{1 - 0}{\frac{\pi}{2} - 0} \Rightarrow \frac{1}{\frac{\pi}{2}} \Rightarrow \frac{2}{\pi} \approx \frac{2}{3.1416} \approx 0.6366$

$\frac{\pi}{2} - 0$

$c = \left(\frac{2}{\pi}\right) \approx 50.46^\circ \approx 0.8806 \text{ rad}$

Ex: 4)  $f(x) = x^2$  on  $[1, 2]$

$\rightarrow$  i)  $f(x) = x^2$  is conti on  $[1, 2]$ .

ii)  $f(x) = x^2$  is diff on  $(1, 2)$ .

iii)  $f'(c) = \frac{f(b) - f(a)}{b - a} \Rightarrow \frac{f(2) - f(1)}{2 - 1} \Rightarrow \frac{4 - 1}{1} \Rightarrow 3$

$b - a$

$2c \Rightarrow 3 \Rightarrow c = 3/2$

verified.

Ex: 5)  $f(x) = x^2 + 1$  on  $[0, 2]$

i)  $f(x) = x^2 + 1$  is conti on  $[0, 2]$

ii)  $f(x) = x^2 + 1$  is diff on  $(0, 2)$

iii)  $f'(c) = \frac{f(b) - f(a)}{b - a}$

$b - a$

$$f(a) = (0)^2 + 1 \Rightarrow 1, \quad f(b) = (2)^2 + 1 \Rightarrow 5$$

$$f(b) - f(a) = 5 - 1 = 4$$

$$f'(c) = \frac{4}{2} = 2$$

$$f(x) = x^2 + 1 \Rightarrow f'(x) = 2x$$

$$f'(c) = 2c = 2 \Rightarrow \boxed{c=1}$$

Ex 10.8.10c

Ex (1) Using LMVT prove that  $\frac{b-a}{\sqrt{1-a^2}} < \sin^{-1}b - \sin^{-1}a$

$$< \frac{b-a}{\sqrt{1-b^2}}$$

→ Let  $f(x) = \sin^{-1}x$ .

- i)  $f(x)$  is cont on  $[a, b]$
- ii)  $f(x)$  is diff on  $(a, b)$
- iii)  $f'(c) = \frac{f(b) - f(a)}{b-a} \quad \text{--- (1)}$

$$f'(c) = \frac{1}{\sqrt{1-c^2}}$$

$$\frac{f(b) - f(a)}{b-a} = \frac{\sin^{-1}b - \sin^{-1}a}{b-a}$$

$$\frac{1}{\sqrt{1-c^2}} = \frac{\sin^{-1}b - \sin^{-1}a}{b-a}$$

Now we have  
 $c \in (a, b)$

$$a < c < b$$

Squaring

$$a^2 < c^2 < b^2$$

$$-a^2 > -c^2 > -b^2$$

$$1-a^2 > 1-c^2 > 1-b^2$$

$$\sqrt{1-a^2} > \sqrt{1-c^2} > \sqrt{1-b^2}$$

reciprocal

$$\frac{1}{\sqrt{1-a^2}} < \frac{1}{\sqrt{1-c^2}} < \frac{1}{\sqrt{1-b^2}}$$

$$\frac{1}{\sqrt{1-a^2}} < \frac{\sin^{-1}b - \sin^{-1}a}{b-a} < \frac{1}{\sqrt{1-b^2}}$$

from (1)

$$\frac{b-a}{\sqrt{1-a^2}} < \sin^{-1}b - \sin^{-1}a < \frac{b-a}{\sqrt{1-b^2}}$$

proved.

2) Using LMVT  $\frac{b-a}{1+b^2} < \tan^{-1}b - \tan^{-1}a < \frac{b-a}{1+a^2}$

→ Let  $f(x) = \tan^{-1}x$ .

- i)  $f(x)$  is cont on  $[a, b]$
- ii)  $f(x)$  is diff on  $(a, b)$
- iii)  $f'(c) = \frac{f(b) - f(a)}{b-a}$

$$f'(c) = \frac{1}{1+c^2}$$

$$\frac{f(b) - f(a)}{b-a} = \frac{\tan^{-1}b - \tan^{-1}a}{b-a}$$

$$\frac{1}{1+c^2} = \frac{\tan^{-1}b - \tan^{-1}a}{b-a} \quad \dots \text{--- (from 1)}$$

$$c \in (a, b)$$

$$a < c < b$$

squaring

$$a^2 < c^2 < b^2$$

$$1+a^2 < 1+c^2 < 1+b^2$$

Reciprocal

$$\frac{1}{1+a^2} > \frac{1}{1+c^2} > \frac{1}{1+b^2}$$

$$\frac{1}{1+a^2} > \frac{\tan^{-1}b - \tan^{-1}a}{b-a} > \frac{1}{1+c^2} \quad \text{--- (from ①)}$$

$$\frac{b-a}{1+a^2} > \tan^{-1}b - \tan^{-1}a > \frac{b-a}{1+c^2}$$

$$\frac{b-a}{1+a^2} > \tan^{-1}b - \tan^{-1}a > \frac{b-a}{1+c^2}$$

Hence proved.

$$3) \text{ Using LMVT prove that } \frac{b-a}{b} < \log\left(\frac{b}{a}\right) < \frac{b-a}{a}$$

→ Let  $f(x) = \log x$ .

i)  $f(x)$  is contn on  $[a, b]$ .

ii)  $f(x)$  is diff on  $(a, b)$ .

$$\text{iii) } f'(c) = \frac{f(b) - f(a)}{b-a}$$

$$f'(c) = 1/c$$

$$\frac{f(b) - f(a)}{b-a} = \log\left(\frac{b}{a}\right)$$

$$\frac{1}{c} = \log\left(\frac{b}{a}\right) = \frac{\log b - \log a}{b-a} \quad \text{--- ①}$$

$$c \in (a, b)$$

$$a < c < b$$

reciprocal.

$$\frac{1}{a} > \frac{1}{c} > \frac{1}{b}$$

$$\frac{1}{a} > \log b - \log a > \frac{1}{b}$$

$$\frac{b-a}{a} > \log b - \log a > \frac{b-a}{b}$$

$$\frac{b-a}{b} < \log b - \log a < \frac{b-a}{a}$$

$$4) \text{ Using LMVT show that } \frac{h}{1+h^2} < \tan^{-1}h < h \text{ on } [0, h]$$

→ i)  $f(x)$  is contn on  $[0, h]$

ii)  $f(x)$  is diff on  $(0, h)$

$$f'(x) = \frac{1}{1+x^2}$$

$$f'(c) = \frac{f(h) - f(0)}{h-0} \Rightarrow \frac{1}{1+c^2} = \frac{\tan^{-1}h - \tan^{-1}0}{h}$$

$$\tan^{-1}0 = 0$$

$$\frac{1}{1+c^2} = \frac{\tan^{-1}h}{h} \quad \text{--- (②)}$$

$$0 < c < h$$

$$0 < c^2 < h^2$$

$$\frac{1}{1+h^2} < \frac{1}{1+c^2} < 1$$

$$\frac{1}{1+h^2} < \frac{1}{1+c^2} < 1 \quad \text{--- squaring.}$$

$$\frac{1}{1+c^2} = \frac{\tan^{-1}h}{h} \text{ put in } \frac{1}{1+h^2} < \frac{\tan^{-1}h}{h} < 1$$

$$\Rightarrow \frac{h}{1+h^2} < \tan^{-1}h < h$$

Hence proved.

## \* Taylor's Theorem

Let  $f(a+h)$  be a func<sup>n</sup> of  $h$  which can be expanded in the power of  $h$ .

$$f(a+h) = f(a) + hf'(a) + \frac{h^2}{2!} f''(a) + \frac{h^3}{3!} f'''(a) + \dots + \frac{h^n}{n!} f^{(n)}(a)$$

$$f'''(a) = \dots + \frac{h^n}{n!} f^{(n)}(a)$$

Ex: Expand  $2x^3 + 7x^2 + x - 6$  in the power of  $x-2$

$$f(x) = 2x^3 + 7x^2 + x - 6$$

$$\text{Let } f(x) = f(x-2+2)$$

Key Step

By Taylor's th<sup>m</sup>

$$f(a+h) = f(a) + hf'(a) + \frac{h^2}{2!} f''(a) + \frac{h^3}{3!} f'''(a)$$

$$f(x-2+2) = f(2) + (x-2)f'(2) + \frac{(x-2)^2}{2!} f''(2) + \frac{(x-2)^3}{3!} f'''(2)$$

$$+ \frac{(x-2)^3}{3!} f'''(2)$$

$$f(2) = 2(2)^3 + 7(2)^2 + 2 - 6$$

$$f'(2) = 6(2)^2 + 14(2) + 1$$

$$f''(2) = 12(2) + 14$$

$$f'''(2) = 12$$

$$f(2) = 2(2)^3 + 7(2)^2 + 2 - 6$$

$$f'(2) = 2(8) + 7(4) + 2 - 6$$

$$f''(2) = 24 + 28 - 6 = 46$$

$$f'''(2) = 38$$

$$f^{(4)}(2) = 12$$

$$\therefore f(x) = 40 + (x-2) \cdot 53 + \frac{(x-2)^2}{2!} \cdot 46 + \frac{(x-2)^3}{3!} \cdot 38$$

$$= 40 + 53(x-2) + 23(x-2)^2 + \frac{19}{3}(x-2)^3$$

Ex: 2

$$x^3 - 2x^2 + 3x + 1 \text{ in power of } x-1$$

$$\text{Let } f(x) = f(x-1+1)$$

$$h = x-1, a=1$$

$$f(x) = f(a) + hf'(a) + \frac{h^2}{2!} f''(a) + \frac{h^3}{3!} f'''(a)$$

$$a=1$$

$$f(x) = x^3 - 2x^2 + 3x + 1$$

$$= (1)^3 - 2(1)^2 + 3(1) + 1 = 1 - 2 + 3 + 1 = 3$$

$$f'(x) = 3x^2 - 4x + 3$$

$$f'(1) = 3(1)^2 - 4(1) + 3 = 3 - 4 + 3 = 2$$

$$f''(1) = 6(1) - 4 = 2$$

$$f'''(1) = 6$$

$$\text{put } f(1) = 3, f'(1) = 2, f''(1) = 2, f'''(1) = 6$$

$$\text{Let } h = (x-1)$$

$$f(x) = 3 + (x-1) \cdot 2 + \frac{(x-1)^2}{2!} \cdot 2 + \frac{(x-1)^3}{3!} \cdot 6$$

$$\implies f(x) = 3 + 2(x-1) + (x-1)^2 + (x-1)^3$$

Ex 3) Expand  $3x^3 - 2x^2 + 2x - 4$  in the power of  $x$

$$\rightarrow \text{let } f'(x+2) \\ h = x+2, \quad 0 = -2$$

$$f(x) = f(0) + f'(0)(x-0) + \frac{f''(0)}{2!}(x-0)^2 + \frac{f'''(0)}{3!}(x-0)^3$$

$$f(-2) = 3(-2)^3 - 2(-2)^2 + (-2) - 4 = -38$$

$$f(x) = 9x^2 - 4x + 1$$

$$f'(-2) = 9(-2) - 4 = -22$$

$$f''(-2) = 9(-2) - 4(-2) + 1 = 36 + 8 + 1 = 45$$

$$f'''(x) = 18x - 4$$

$$f'''(-2) = 18(-2) - 4 = -36 - 4 = -40$$

$$\frac{f'''(-2)}{2!} = \frac{-40}{2} = -20$$

$$f'''(x) = 18$$

$$\frac{f'''(-2)}{3!} = \frac{18}{6} = 3$$

$$3(x+2)^3 - 20(x+2)^2 + 45(x+2) - 38$$

Ex 3)

Using Taylor's theorem, express  $(x-2)^4 - 3(x-2)^3 + 4(x-2)^2 + 5$  in power of  $x$ .

$$\rightarrow f(x) - f(x-2) = f(x-2)$$

$$h = x-2$$

$$f(x-2) = f(0) + hf'(0) + \frac{h^2}{2!}f''(0) + \frac{h^3}{3!}f'''(0) + \frac{h^4}{4!}f^{(4)}(0)$$

$$f'''(0) = 12$$

$$= f(-2) + 2f'(-2) + \frac{2^2}{2!}f''(-2) + \frac{2^3}{3!}f'''(-2) + \frac{2^4}{4!}f^{(4)}(-2)$$

$$\frac{2^3}{3!}f'''(-2) + \frac{2^4}{4!}f^{(4)}(-2)$$

$$\text{put } x-2 = t$$

$$f(t) = t^4 - 3t^3 + 4t^2 + 5$$

$$f'(t) = 4t^3 - 9t^2 + 8t$$

$$f''(t) = 12t^2 - 18t + 8$$

$$f'''(t) = 24t - 18$$

$$f^{(4)}(t) = 24$$

$$f(-2) = 61$$

$$f'(-2) = -84$$

$$f''(-2) = 92$$

$$f'''(-2) = -66$$

$$f^{(4)}(-2) = 24$$

$$\Rightarrow x^4 - 11x^3 + 46x^2 - 84x + 61$$

$$\Rightarrow x^4 - 11x^3 + 46x^2 - 84x + 61$$

\* Maclaurin's theorem:

$$\text{put } a=0, h=x$$

$$f(x) = f(0) + x f'(0) + \frac{x^2}{2!} f''(0) + \frac{x^3}{3!} f'''(0) + \dots$$

$$\frac{x^4}{4!} f^{(4)}(0) + \dots$$

Q.1) Expand  $e^x$  by Maclaurin's Th<sup>m</sup> upto  $x^5$ .

$$\rightarrow f(x) = f(0) + x f'(0) + \frac{x^2}{2!} f''(0) + \frac{x^3}{3!} f'''(0) + \dots$$

$$\frac{x^4}{4!} f^{(4)}(0) + \frac{x^5}{5!} f^{(5)}(0) + \dots$$

$$f(x) = e^x, \quad f(0) = e^0 = 1$$

$$f'(x) = e^x, \quad f'(0) = e^0 = 1$$

$$f''(x) = e^x, \quad f''(0) = e^0 = 1$$

$$f'''(x) = e^x, \quad f'''(0) = e^0 = 1$$

$$f^{(4)}(x) = e^x, \quad f^{(4)}(0) = e^0 = 1$$

$$f^{(5)}(x) = e^x, \quad f^{(5)}(0) = e^0 = 1$$

$$f(x) = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \frac{x^5}{5!} + \dots$$

$$2! = 1 \times 2 = 2$$

$$3! = 1 \times 2 \times 3 = 6$$

$$4! = 1 \times 2 \times 3 \times 4 = 24$$

$$5! = 1 \times 2 \times 3 \times 4 \times 5 = 120$$

2) Expand  $\sin x$  by Maclaurin's Th<sup>m</sup> upto  $x^5$ .

$$\rightarrow f(x) = f(0) + x f'(0) + \frac{x^2}{2!} f''(0) + \frac{x^3}{3!} f'''(0) + \dots$$

$$\frac{x^4}{4!} f^{(4)}(0) + \frac{x^5}{5!} f^{(5)}(0) + \dots$$

$$f(x) = \sin x, \quad f(0) = \sin 0 = 0$$

$$f'(x) = \cos x, \quad f'(0) = \cos 0 = 1$$

$$f''(x) = -\sin x, \quad f''(0) = -\sin 0 = 0$$

$$f'''(x) = -\cos x, \quad f'''(0) = -\cos 0 = -1$$

$$f^{(4)}(x) = \sin x, \quad f^{(4)}(0) = \sin 0 = 0$$

$$f^{(5)}(x) = \cos x, \quad f^{(5)}(0) = \cos 0 = 1$$

$$f(x) = 0 + x + \frac{x^3}{3!} (-1) + \frac{x^5}{5!} (1) + \dots$$

$$\frac{x^5}{5!} (1)$$

$$\Rightarrow f(x) = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots$$

$$\text{Ans: } \sin x = x - \frac{x^3}{6} + \frac{x^5}{120} - \dots$$

3) Expand  $\tan x$  by Maclaurin's th<sup>m</sup> upto  $x^3$

$$\rightarrow f(x) = f(0) + xf'(0) + \frac{x^2}{2!} f''(0) + \frac{x^3}{3!} f'''(0)$$

$$f(x) = \tan x$$

$$f'(x) = \sec^2 x$$

$$f''(x) = 2(\sec x) \cdot (\sec x \tan x) = 2 \sec^2 x \tan x$$

$$f'''(x) = 2 \left[ \sec x \cdot \sec^2 x + \tan x \cdot \sec^2 x \right]$$

$$= 2 \left[ \sec^3 x + \sec^2 x \tan x \right]$$

$$f(0) = \tan(0) = 0$$

$$f'(0) = \sec^2(0) = 1^2 = 1$$

$$f''(0) = 2 \sec^2(0) \tan(0) = 2(1)^2(0) = 0$$

$$f'''(0) = 2 \sec^3(0) + 4 \sec^2(0) \tan(0) = 2(1)^3 + 4(1)^2(0) = 2$$

$$f(x) = f(0) + xf'(0) + \frac{x^2}{2!} f''(0) + \frac{x^3}{3!} f'''(0)$$

$$\tan x = 0 + x(1) + \frac{x^2}{2} (0) + \frac{x^3}{6} (2)$$

$$\tan x = x + \frac{2}{6} x^3$$

$$\tan x = x + \frac{1}{3} x^3$$

$$\tan x \approx x + \frac{1}{3} x^3$$

4) Expand  $f(x) = \sec x$  upto  $x^3$

$$\rightarrow f(x) = f(0) + xf'(0) + \frac{x^2}{2!} f''(0) + \frac{x^3}{3!} f'''(0) + \dots$$

$$f(x) = \sec x$$

$$f'(x) = \sec x \tan x$$

$$f''(x) = \sec x \cdot \frac{d}{dx} (\tan x) + \tan x \cdot \frac{d}{dx} (\sec x)$$

$$f''(x) = \sec x (\sec^2 x) + \tan x (\sec x \tan x)$$

$$f''(x) = \sec^3 x + \sec x \tan^2 x$$

$$f'''(x) = \sec^3 x + \sec x (\sec^2 x - 1) = 2 \sec^3 x - \sec x$$

$$f'''(x) = 6 \sec^3 x \tan x - \sec x \tan x$$

$$f(0) = \sec(0) = 1$$

$$f'(0) = \sec(0) \tan(0) = 1 \cdot 0 = 0$$

$$f''(0) = 2 \sec^3(0) - \sec(0) = 2(1)^3 - 1 = 1$$

$$f'''(0) = 6(1)^3(0) - (1)(0) = 0$$

$$f(x) = 1 + \frac{x^2}{2!} (1) + \frac{x^3}{3!} (0)$$

$$f(x) = 1 + \frac{x^2}{2}$$

Q4) Q3) 1/5

Indeterminant form:

$$\frac{0}{0}, \frac{\infty}{\infty}, \infty \cdot 0, 0 \cdot \infty, \infty - \infty, 0 \times \infty$$

1) Type 1:  $0/0$  form.

ex) Evaluate  $\lim_{x \rightarrow 0} \frac{e^{ax} - e^{-ax}}{\log(1+bx)}$

$$\rightarrow \text{put } x=0. \quad \frac{e^0 - e^0}{\log(1+0)} = \frac{1-1}{0} = \frac{0}{0} \text{ form.}$$

$\therefore$  Apply LHR

$$\lim_{x \rightarrow 0} \frac{\frac{d}{dx}(e^x - e^{-x})}{\frac{d}{dx}(\log(1+bx))}$$

$$\lim_{x \rightarrow 0} \frac{e^{0x} - (-e^{-0x})}{\frac{1}{1+bx} (b)}$$

$$\lim_{x \rightarrow 0} \frac{e^{0x} + e^{-0x}}{\frac{b}{1+bx}}$$

$$\text{put } x=0. \quad \therefore \frac{e^0 + e^0}{\frac{b}{1+0}} = \frac{a(1) + a(1)}{b} = \frac{2a}{b}$$

2) Type 2:  $\infty/\infty$

ex; Evaluate  $\lim_{x \rightarrow 0} \frac{\log \tan x}{\log x}$

$$\rightarrow \text{put } x \Rightarrow 0. \quad \frac{\log \tan 0}{\log(0)} = \frac{\infty}{\infty}$$

$\therefore$  Apply LHR

$$\lim_{x \rightarrow 0} \frac{\frac{d}{dx}(\log \tan x)}{\frac{d}{dx} \log x}$$

$$\text{let } \tan x = x. \quad \lim_{x \rightarrow 0} \frac{1}{\tan x} \cdot \sec^2 x$$

$$\lim_{x \rightarrow 0} x (\cot x \cdot \sec^2 x)$$

$$\lim_{x \rightarrow 0} x \left( \frac{\cos x}{\sin x} \cdot \frac{1}{\cos^2 x} \right) \Rightarrow \lim_{x \rightarrow 0} \frac{x}{\sin x \cos x}$$

$$\lim_{x \rightarrow 0} \frac{\sec x}{\sin x} \text{ put } x=0.$$

$$\therefore \lim_{x \rightarrow 0} \frac{x}{\sin x} = 1$$

$$\frac{1}{\cos 0} = 1$$

ex; 2) Evaluate  $\lim_{x \rightarrow 0} \log \tan^4 x$

$$\rightarrow \log A = \log B$$

$$\lim_{x \rightarrow 0} \log \tan^4 x, \text{ put } x=0 \Rightarrow \log \tan^4 0 = \frac{\infty}{\infty}$$

$$\text{Apply LHR.} \quad \lim_{x \rightarrow 0} \frac{\frac{d}{dx}(\log \tan^4 x)}{\frac{d}{dx}(\log \tan^4 x)} \Rightarrow \lim_{x \rightarrow 0} \frac{4 \sec^2 x / \tan^4 x}{\sec^2 x / \tan^4 x}$$

$$\Rightarrow \lim_{x \rightarrow 0} \left( \frac{4 \sec^2(4x)}{\sec^2(x)} \cdot \frac{\tan x}{\tan^4 x} \right)$$

$$\Rightarrow \lim_{x \rightarrow 0} \frac{4 \sec^2(4x)}{\sec^2(x)} = \frac{4(1)^2}{(1)^2} = 4$$

$$\Rightarrow \lim_{\theta \rightarrow 0} \frac{\tan \theta}{\theta} = 1$$

$$\lim_{x \rightarrow 0} \frac{\tan x}{\tan 4x} = \lim_{x \rightarrow 0} \left( \frac{\frac{\tan x}{x}}{\frac{\tan 4x}{4x}} \right) \cdot \frac{4x}{x} = \frac{1}{4} \cdot \lim_{x \rightarrow 0} \frac{\frac{\tan x}{x}}{\frac{\tan 4x}{4x}}$$

$$L = 4 \times \frac{1}{4} = 1$$

Observe

5) Type  $-0^0, \infty^0, 1^\infty$

ex: 1) Evaluate  $\lim_{x \rightarrow \pi/2} (\cos x)^{\cos x}$

→ put  $x = \pi/2$   
 $\cos 90^\circ = 0^0$   
 $\left( \cos \frac{\pi}{2} \right)^{\cos \pi/2} = 0^0$

Let  $L = \lim_{x \rightarrow \pi/2} \cos x^{\cos x}$

Taking logs both side  
 $\log L = \lim_{x \rightarrow \pi/2} \cos x \log \cos x$

put  $x = \pi/2$   
 $\cos \frac{\pi}{2} \times \log \cos \frac{\pi}{2}$

$0 \times \infty \Rightarrow$  form

$\log L = \lim_{x \rightarrow \pi/2} \frac{\log \cos x}{\frac{1}{\cos x}}$

$\log L = \lim_{x \rightarrow \pi/2} \frac{\log \cos x}{\sec x}$

put  $x = \frac{\pi}{2}$

$\frac{0}{0}$  form.

Apply LHR  
 $\lim_{x \rightarrow \pi/2} \frac{\frac{d}{dx} \log \cos x}{\frac{d}{dx} \sec x} \Rightarrow \log \lim_{x \rightarrow \pi/2} \frac{\frac{1}{\cos x} (-\sin x)}{\sec x \cdot \tan x}$

$\log L = \lim_{x \rightarrow \pi/2} \frac{-\cancel{\sin x} \cdot \cos x}{\sec x \cdot \cancel{\sin x}}$

$\log L = \lim_{x \rightarrow \pi/2} (-\cos x)$

$\log L = 0 \Rightarrow L = e^0 \Rightarrow \boxed{L = 1}$

2) Evaluate  $\lim_{x \rightarrow 0} (\cot x)^{\sin x}$

→ put  $x = 0$   
 $\cot 0^\circ = \infty, \sin 0^\circ = 0$   
 $(\cot x)^{\sin x} = \infty^0$

Let  $L = \lim_{x \rightarrow 0} (\cot x)^{\sin x}$

Taking logs both sides  
 $\ln L = \lim_{x \rightarrow 0} \ln ((\cot x)^{\sin x})$

$\ln L = \lim_{x \rightarrow 0} \sin x \ln (\cot x) = \lim_{x \rightarrow 0} (\ln(\infty) \cdot 0)$

L' Hospital's rule  
 $\ln L = \lim_{x \rightarrow 0} \frac{\ln (\cot x)}{\csc x} = \frac{0}{\infty}$

$\frac{d}{dx} [\ln (\cot x)] = \frac{1}{\cot x} \cdot (-\csc^2 x) = -\sin x \cdot \frac{1}{\cos x \sin^2 x}$

$= -\frac{1}{\sin \cos x}$

$\frac{d}{dx} [\csc x] = -\csc x \cot x = -\frac{1}{\sin x} \cdot \frac{\cos x}{\sin x} = -\frac{\cos x}{\sin^2 x}$

$$\ln L = \lim_{x \rightarrow 0} \frac{1}{\frac{\sin x \cos x}{\cot x}} = \frac{1}{\frac{\sin x \cos x}{\cot x}}$$

$$\ln L = \lim_{x \rightarrow 0} \left( \frac{1}{\sin x \cos x} \cdot \frac{\sin^2 x}{\cos x} \right)$$

$$\ln L = \lim_{x \rightarrow 0} \frac{\sin x}{\cos^2 x}$$

$$\text{put } x = 0$$

$$\ln L = \frac{\sin(0)}{\cos^2(0)} = \frac{0}{1} = 0$$

$$L = e^0 = 1$$

$$\text{Ans: } \lim_{x \rightarrow 0} (\cot x)^{\sin x} = 1$$

3) Evaluate  $\lim_{x \rightarrow 0} (1 + \tan x)^{\cot x}$ .

$$\text{put } x = 0$$

$$\tan 0 = 0, \cot 0 = \infty$$

$$(1 + \tan x)^{\cot x} = (1 + 0)^{\infty} = (1)^{\infty}$$

$$\text{let } L = \lim_{x \rightarrow 0} (1 + \tan x)^{\cot x}$$

Taking logs both sides.

$$\log L = \lim_{x \rightarrow 0} (1 + \tan x) \log(\cot x)$$

$$\ln L = \lim_{x \rightarrow 0} \ln[(1 + \tan x)^{\cot x}]$$

$$\ln(a^b) = b \ln(a)$$

$$\ln L = \lim_{x \rightarrow 0} \cot x \cdot \ln(1 + \tan x)$$

$$\cot x = \frac{1}{\tan x}$$

$$\ln L = \lim_{x \rightarrow 0} \ln(1 + \tan x)$$

$$\lim_{u \rightarrow 0} \ln(1 + u) = 1 \cdot \tan x \quad \text{formula}$$

$$\ln L = 1$$

$$L = e^1 = e$$

$$\text{Ans: } \lim_{x \rightarrow 0} (1 + \tan x)^{\cot x} = e$$

4)  $\lim_{x \rightarrow 0} (\sin x)^{\tan x}$

$$\text{put } x = 0$$

$$\sin 0 = 0, \tan 0 = 0$$

$$(\sin x)^{\tan x} = 0^0$$

$$\text{let } L = \lim_{x \rightarrow 0} (\sin x)^{\tan x} = 0^0$$

$$\ln L = \lim_{x \rightarrow 0} \ln[(\sin x)^{\tan x}]$$

$$\ln L = \lim_{x \rightarrow 0} \tan x \ln(\sin x)$$

$$[0 \times (-\infty)] \text{ form}$$

Express  $\tan x$  as  $1/\cot x$  to turn it into  $\frac{\infty}{\infty}$ .

$$\ln L = \lim_{x \rightarrow 0} \ln(\sin x)$$

$$\text{Numerator: } \frac{d}{dx} [\ln(\sin x)] = \frac{\cos x}{\sin x} = \cot x$$

$$\text{Denominator: } \frac{d}{dx} [\cot x] = -\csc^2 x$$

$$\ln L = \lim_{x \rightarrow 0} \frac{\cot x}{-\csc^2 x}$$

$$\ln L = \lim_{x \rightarrow 0} \frac{\cos x / \sin x}{-1 / \sin^2 x}$$

$$\ln L = \lim_{x \rightarrow 0} (-\cos x \cdot \sin x)$$

put  $x = 0$ .

$$\ln L = -\cos(0) \cdot \sin(0) = -1 \cdot 0 = 0.$$

$$\Rightarrow \ln L = 0$$

$$\Rightarrow L = e^0$$

## Unit 2

### Fourier Series:

Definition: Let  $f(x)$  is periodic function define in the interval  $(0, 2\pi)$   $0 \leq x \leq 2\pi$ .

Fourier series can be expressed as  $f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)$

$$\text{i.e. } f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx + \sum_{n=1}^{\infty} b_n \sin nx$$

$$a_0 = \frac{1}{\pi} \int_0^{2\pi} f(x) dx.$$

$$a_n = \frac{1}{\pi} \int_0^{2\pi} f(x) \cos nx dx.$$

$$b_n = \frac{1}{\pi} \int_0^{2\pi} f(x) \sin nx dx.$$

$$f(x) = \sin n.$$

$$f(-x) = \sin(-x) \Rightarrow -\sin x.$$

$$f(x) = f(x) \rightarrow \text{even func.}$$

$$f(-x) = -f(x) \rightarrow \text{odd func.}$$

$$\text{e.g. } f(x) = x^2, \quad f(x) = x.$$

$$f(-x) = (-x)^2 = x^2 \rightarrow \text{even func.}$$

$$f(-x) = -x \rightarrow \text{odd func.}$$

$$\text{eg: } f(x) = \pi^2 - x^2$$

$$f(-x) = \pi^2 - (-x)^2$$

$$= \pi^2 - x^2$$

\* Bernoulli's theorem:

$$\int uv = uv_1 - u'v_2 + u''v_3 - u'''v_4 + \dots + (-1)^n u^{(n)}v_{n+1}$$

standard formula for UV.

| $0 \leq x \leq 2\pi$                                | $-\pi \leq x \leq \pi$                               | $0 \leq x \leq 2L$                                            | $-L \leq x \leq L$                                            | Half range cosine        | Half range sine         |
|-----------------------------------------------------|------------------------------------------------------|---------------------------------------------------------------|---------------------------------------------------------------|--------------------------|-------------------------|
| $a_0 = \frac{1}{\pi} \int_0^{2\pi} f(x) dx$         | odd or even<br>If $f(x)$ even<br>func <sup>n</sup> . | $a_0 = \frac{1}{L} \int_0^{2L} f(x) dx$                       | odd or even.                                                  | even func <sup>n</sup> . | odd func <sup>n</sup> . |
| $a_n = \frac{1}{\pi} \int_0^{2\pi} f(x) \cos nx dx$ | $b_n = 0$<br>To find $a_0, a_n$ .                    | $a_n = \frac{1}{L} \int_0^{2L} f(x) \cos \frac{n\pi x}{L} dx$ | $-1 \leq x \leq 1$<br>eg: $x$                                 | $b_n = 0$                | $a_0, a_n = 0$          |
| $b_n = \frac{1}{\pi} \int_0^{2\pi} f(x) \sin nx dx$ | * $f(x)$ odd.<br>$a_0, a_n = 0$<br>To find $b_n$ .   | $b_n = \frac{1}{L} \int_0^{2L} f(x) \sin \frac{n\pi x}{L} dx$ | $0 \leq x \leq \pi$<br>$0 \leq x \leq L$<br>eg: $x, x^2, x^3$ |                          |                         |

★ Bernoulli's theorem :

$$\int uv = uv_1 - u'v_2 + u''v_3 - u'''v_4 + \dots + (-1)^n u^{(n)}v_{n+1}$$

Standard formula for UV.

| $0 \leq x \leq 2\pi$                                      | $-\pi \leq x \leq \pi$                               | $0 \leq x \leq 2L$                                                  | $-L \leq x \leq L$                        | Half range cosine                                             | Half range sine                          |
|-----------------------------------------------------------|------------------------------------------------------|---------------------------------------------------------------------|-------------------------------------------|---------------------------------------------------------------|------------------------------------------|
| $a_0 = \frac{1}{\pi} \int_0^{2\pi} f(x) dx$               | odd or even<br>If $f(x)$ even<br>func <sup>n</sup> . | $a_0 = \frac{1}{L} \int_0^{2L} f(x) dx$                             | odd or even.<br>$-1 \leq x \leq 1$<br>ex; | even<br>func <sup>n</sup> .                                   | odd<br>func <sup>n</sup> .               |
| $a_n = \frac{1}{\pi} \int_0^{2\pi} f(x) \cdot \cos nx dx$ | $b_n = 0$<br>To find<br>$a_0, a_n$ .                 | $a_n = \frac{1}{L} \int_0^{2L} f(x) \cdot \cos \frac{n\pi x}{L} dx$ |                                           | $b_n = 0$                                                     | $a_0, a_n = 0$                           |
| $b_n = \frac{1}{\pi} \int_0^{2\pi} f(x) \cdot \sin nx dx$ | * $f(x)$ odd.<br>$a_0, a_n = 0$<br>To find $b_n$ .   | $b_n = \frac{1}{L} \int_0^{2L} f(x) \cdot \sin \frac{n\pi x}{L} dx$ |                                           | $0 \leq x \leq \pi$<br>$0 \leq x \leq L$<br>eg: $x, x^2, x^3$ | $0 \leq x \leq \pi$<br>$0 \leq x \leq L$ |